

Cloud Computing

Where do your Google Photos actually live?

Where does Netflix store thousands of movies for millions of users?

And how can you edit a document with your friend at the same time on Google Docs?

Cloud computing is the delivery of computing services over the internet, including storage, processing power, databases, networking, and more.

Cloud computing means **renting computers instead of buying them** — over the internet!

Key Characteristics :

- **On-demand self-service:** Users can provision computing resources (e.g., servers, storage, databases) automatically without requiring human interaction with the service provider.
- **Resource pooling:** Provider's computing resources (e.g., servers, storage) are pooled and shared among multiple users dynamically. Resources are allocated and reallocated based on demand.
- **Rapid elasticity:** The ability to scale resources up or down quickly and automatically based on demand.
- **Scalability:** The ability to increase or decrease resources as needed, either manually or automatically.

Vertical Scaling : Increasing the capacity of a single resource (e.g., adding more CPU or memory to a VM).

Horizontal Scaling : Adding more resources (e.g., launching additional VMs).

- **Pay-As-You-Go:** Users pay only for the resources they consume, rather than purchasing and maintaining physical hardware.
- **Global Reach:** Cloud services are available worldwide, with data centers located in multiple regions to reduce latency and improve performance.
- **Security:** Cloud providers implement robust security measures to protect data and applications, including encryption, firewalls, and identity management.

Benefits :

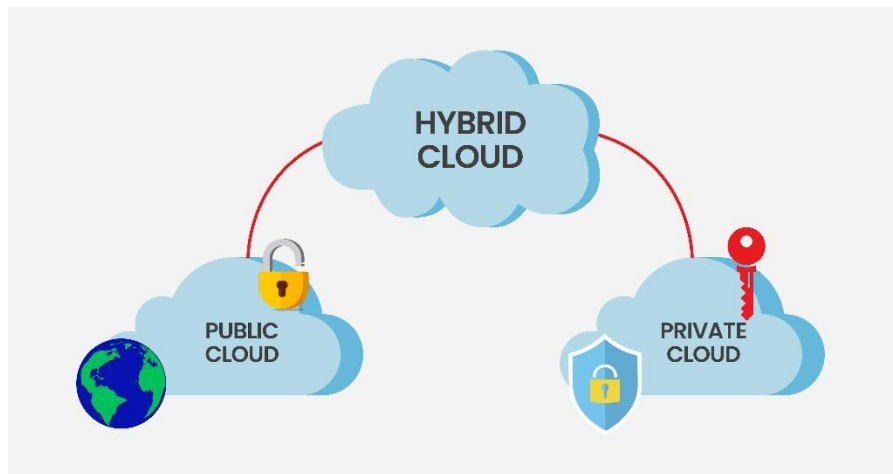
- Cost savings (pay-as-you-go model).
- Scalability and flexibility.
- High availability and reliability.
- Global reach.

Challenges :

- Security concerns.
- Dependency on internet connectivity.
- Vendor lock-in.
- Data privacy issues.

Types of Cloud Deployment Models :

- Public Cloud (e.g., AWS, Microsoft Azure).
- Private Cloud (dedicated infrastructure for a single organization).
- Hybrid Cloud (combination of public and private clouds).

**Types of Cloud Service Models :****1. IaaS (Infrastructure as a Service):**

Rent servers, storage, and networks from a cloud provider. Users manage the OS, apps, and security, while the vendor handles hardware. Offers high flexibility but requires more maintenance effort.

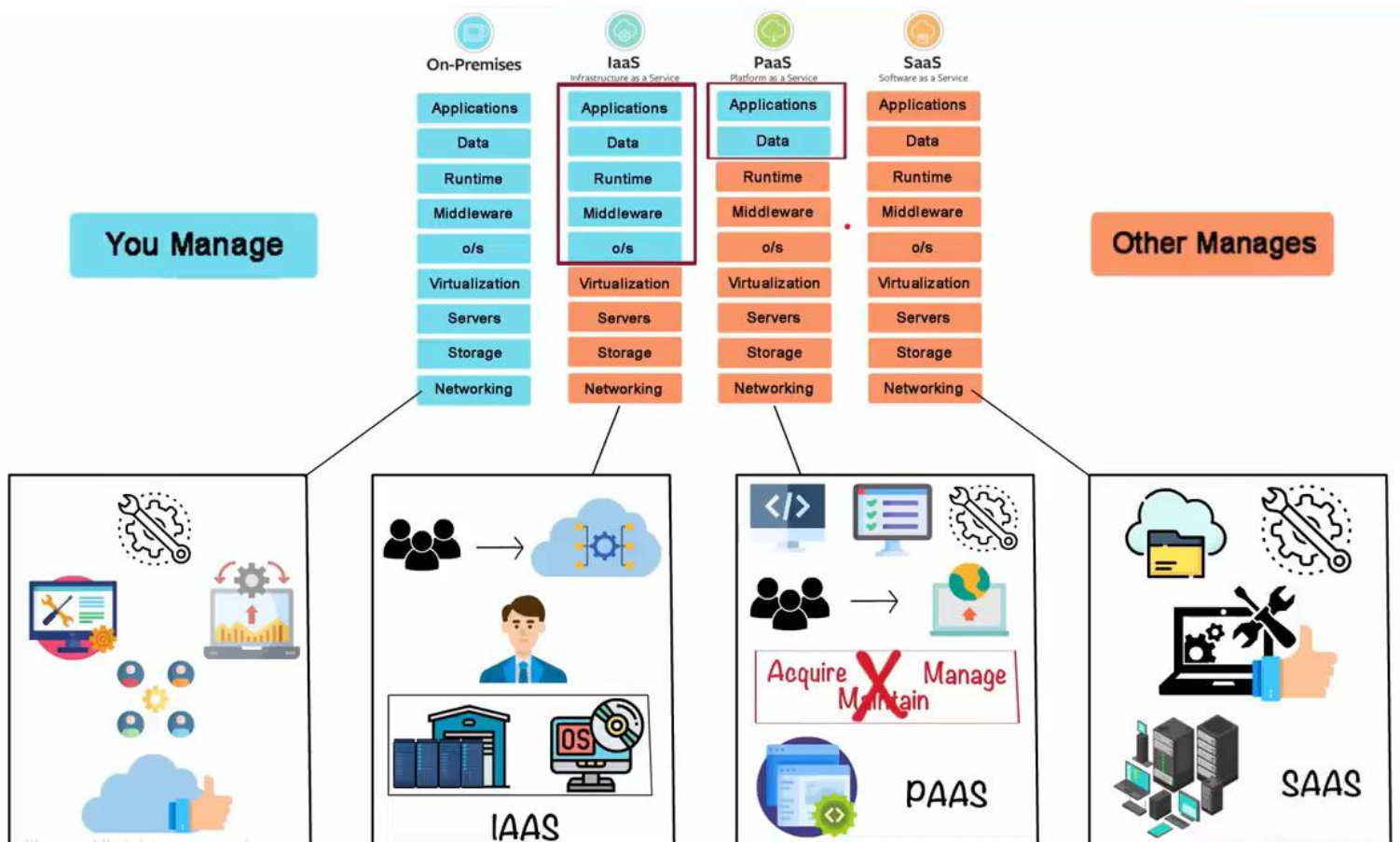
2. PaaS (Platform as a Service):

Provides an environment for developing, testing, and deploying apps. Developers

focus on coding, while the vendor manages the infrastructure, reducing maintenance but limiting flexibility.

3. SaaS (Software as a Service):

Delivers ready-to-use software over the internet on a subscription basis (e.g., Dropbox, Office 365). Fully managed by the vendor, it minimizes costs and effort for users.



Q: In which one of the following deployment models are you, as the business, responsible for the Application, Data and Operating System?

1. IaaS
2. PaaS
3. SaaS

Q: Name the most popular cloud computing services in the market

AWS, Azure, Google cloud platform

Q: Can you name three things you use daily that might be running on the cloud without you realizing it?

Purpose : To connect cloud computing to everyday life (e.g., email services, Office 365, Google Drive).